Gosset-KEM: E8 Message Coding for ML-KEM and Correlation-Based Failure Boosting against Multi-Dimensional Lattice Codes

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Abstract

Replacing the one-bit-per-coefficient message encoding of ML-KEM (FIPS 203) by a multi-dimensional lattice code lowers the decryption failure rate (DFR), and the margin can be spent on smaller ciphertexts or on more noise. We study the Gosset lattice E_8 in this role and make four contributions. (i) We give exact DFRs for all three ML-KEM levels and validate the independence assumption behind them on real ring encryptions. (ii) We define two families of parameter sets with FIPS 203 key sizes: *Compact*, with 6–10% smaller ciphertexts at unchanged lattice security, and *MaxSec*, which gains +5 to +11 bits against the strongest attacks of the lattice-estimator (dual-hybrid, MATZOV cost model) at unchanged sizes. (iii) We identify a *correlation-based failure-boosting attack* specific to multi-dimensional codes. Because an E_8 block fails along directions that combine 8 coefficients, an adversary who grinds ciphertexts with a large autocorrelation of the encryption randomness at the in-block index distances raises their failure probability; the standard decoder is immune. With consecutive blocks, the cost of a first failure within 2^{64} queries drops to 2^{127} for the level-1 Compact set, below its claimed security. We propose a public coefficient layout in which the 28 pairwise index distances of every block are distinct. It raises the cost to 2^{192}, leaves the average DFR unchanged, and costs nothing. We show that the Leech lattice is inherently more exposed. (iv) We provide a complete constant-time C implementation (NIST ACVP vectors pass), a first-order masked decapsulation with a bit-sliced masked E_8 decoder, a hybrid mode with X25519, and an analysis of compiler-induced timing leaks found on two compilers and two CPU vendors. We also report a negative result: placing E_8 structure in the LWE *noise* instead of the message weakens security by $\log_2$ of the sublattice index.

1 Introduction

ML-KEM [1] encodes each message bit into one polynomial coefficient as 0 or $\lceil q/2 \rfloor$ and decodes by rounding. Lattice codes improve this: they pack message points densely in several dimensions, so a larger noise vector is tolerated for the same failure probability [4, 3, 5, 2]. Liu and Sakzad recently proposed E_8-type codes for Kyber [2], reporting a DFR of 2^{-286} for Kyber-1024 under a Gaussian approximation. We call our family *Gosset-KEM*, after Thorold Gosset, who in 1900 discovered the 4_{21} polytope whose 240 vertices are the minimal vectors of E_8[1].

Previous work treats the DFR of a lattice-coded scheme as an *average* over honest ciphertexts. In the chosen-ciphertext setting, however, an adversary chooses which ciphertexts to

[1]Not to be confused with W. S. Gosset ("Student"). The project was developed under the working name AFA-12.

submit, and failure-boosting attacks [6, 7] exploit ciphertexts with a higher-than-average failure probability. This paper asks whether the multi-dimensional structure of the code itself gives the adversary new ways to find such ciphertexts. It does.

Contributions.

1. **Exact DFRs and validation** (§4). By exact convolution with a union bound over the 240 Voronoi-relevant vectors of E_8 we obtain $2^{-240.7}/2^{-279.4}/2^{-335.0}$ at the ML-KEM-512/768/1024 parameters (standard decoding: $2^{-139.1}/2^{-165.3}/2^{-175.2}$). The Gaussian approximation reproduces [2] but underestimates the gain. The independence assumption is confirmed with real encryptions at inflated noise: measured $1.711 \cdot 10^{-5}$ against predicted $1.705 \cdot 10^{-5}$.
2. **Parameter sets** (§5): Compact and MaxSec, evaluated with the lattice-estimator [9] on both the key and the ciphertext LWE instances.
3. **A new attack and countermeasure** (§6). This is our main contribution: correlation-based failure boosting, a rigorous tail model (Chernoff bound and Lugannani–Rice saddlepoint, validated against Monte Carlo), the distinct-distance layout, directional boosting after the first failure, and the Leech lattice.
4. **Implementation** (§7): constant-time decoding, masking, a hybrid mode, cross-platform tests, and the compiler and microarchitecture findings.
5. **A negative result** (§3): E_8-structured noise.

All code, test vectors and analysis scripts are public, and every number in this paper is reproducible from them.

2 Background

ML-KEM. Let $q = 3329$, $n = 256$, $R_q = \mathbb{Z}_q[X]/(X^n + 1)$. K-PKE has secret $\mathbf{s}$, error $\mathbf{e} \leftarrow \beta_{\eta_1}^k$ and public key $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$. Encryption samples $\mathbf{r} \leftarrow \beta_{\eta_1}^k$, $\mathbf{e}_1 \leftarrow \beta_{\eta_2}^k$, $e_2 \leftarrow \beta_{\eta_2}$ from $G(m\|H(\mathrm{ek}))$ and outputs $\mathbf{u} = \mathrm{Compress}_{d_u}(\mathbf{A}^T\mathbf{r} + \mathbf{e}_1)$ and $v = \mathrm{Compress}_{d_v}(\mathbf{t}^T\mathbf{r} + e_2 + \mu)$. Decryption computes

$$w = v - \mathbf{s}^T\mathbf{u} = \mu + \underbrace{\mathbf{e}^T\mathbf{r} - \mathbf{s}^T(\mathbf{e}_1 + \mathbf{c}_u) + e_2 + c_v}_{\delta},$$

where $\mathbf{c}_u, c_v$ are the compression errors. The Fujisaki–Okamoto transform with implicit rejection turns K-PKE into an IND-CCA2 KEM; its security bound contains the DFR [8].

Decryption errors as inner products. Write $S = (\mathbf{e}, -\mathbf{s}) \in \mathbb{Z}^{2kn}$ (secret) and $C = (\mathbf{r}, \mathbf{e}_1 + \mathbf{c}_u)$, $G = e_2 + c_v$. These are known to whoever generates the ciphertext, which in the CCA setting means anyone who chooses m. Then $\delta_i = \langle S, C_i \rangle + G_i$, where C_i is the negacyclic rotation of C that produces coefficient i.

E_8. $\mathsf{E}_8 = D_8 \cup (D_8 + \frac{1}{2}\mathbf{1})$ with $D_8 = \{x \in \mathbb{Z}^8 : \sum x_i \equiv 0 \pmod 2\}$. It has 240 minimal vectors of squared norm 2, which are also its Voronoi-relevant vectors: 112 of type $(\pm 1^2, 0^6)$ and 128 of type $(\pm\frac{1}{2}^8)$ with an even number of minus signs. Nearest-point decoding uses two D_8 decoders [11]. The code $(q/2)\mathsf{E}_8$ mod $q\mathbb{Z}^8$ carries 8 bits per 8 coefficients, the same rate as ML-KEM.

3 A negative result: E_8 in the noise

Our starting idea was to sample LWE errors from a structured distribution over E_8 (Construction A of the $[8, 4, 4]$ Hamming code). It fails.

Observation 1. *If the error vector lies in a sublattice $L \subset \mathbb{Z}^k$ of index D, an attacker who knows L can reduce LWE with error σ to an instance with error $\sigma D^{-1/k}$ and loses nothing. Equivalently, the entropy drops by $\log_2 D$ bits per block. For integral E_8 ($\sqrt{2}\mathsf{E}_8$, $D = 16$) this halves the effective variance.*

With a primal-attack estimator validated against Kyber, equal-variance E_8 noise lost 5, 11, 17 and 22 bits at $n = 256, 512, 768, 1024$. Real LLL/BKZ attacks on small instances confirmed it: a structure-aware attacker succeeded in 60% of trials, against 0% with i.i.d. noise of the same variance, matching the behaviour of i.i.d. noise at half the variance. Lattice codes belong in the *message*: they protect, they do not hide.

4 E_8 message coding: exact DFR and validation

Encoding and decoding. Byte b of m gives a coset bit c and $x_0..x_6$, with $x_7 = \oplus x_i$. The coefficients are $\mu_i = \mathrm{round}((q/2)(x_i + c/2))$. Decoding is nearest-point decoding in $(q/2)\mathsf{E}_8$ using integer arithmetic on $Y = 4\,\mathrm{cent}(w)$, with deterministic tie-breaking; the specification is in the repository.

Exact DFR. We compute the exact distribution of each coefficient of δ by convolution, which reproduces the official ML-KEM DFRs exactly. Assuming the 8 coefficients of a block are independent, block failure is bounded by the union over the 240 relevant vectors u of $\Pr[\langle \delta, u \rangle > q/2]$:

$\log_2$ DFR	ML-KEM-512	ML-KEM-768	ML-KEM-1024
standard decoding	-139.1	-165.3	-175.2
E_8, exact	-240.7	-279.4	-335.0
E_8, Gaussian approximation	-152.2	-162.7	-290.9

The Gaussian approximation is very pessimistic when $d_v = 4$, because the compression error is bounded and not Gaussian. It reproduces the 2^{-286} of [2] within 5 bits.

Validation of independence. In a real ciphertext the 8 coefficients of a block share $\mathbf{s}, \mathbf{e}$. We ran real ring encryptions at inflated noise ($k = 2$, $\eta = 13$) to make failures measurable. The block failure rate was $1.711 \cdot 10^{-5}$ against $1.705 \cdot 10^{-5}$ predicted, and the independent C implementation measured $2.0 \cdot 10^{-5}$ over $4.8 \cdot 10^{6}$ blocks. On average, the independence model is accurate. Section 6 shows that this does *not* hold for adversarially chosen ciphertexts.

5 Parameter sets

set	k	η_1	η_2	d_u	d_v	ct (B)	$\log_2$ DFR
Gosset-KEM-512-Compact	2	3	2	9	4	704 (-8.3%)	-142.0
Gosset-KEM-768-Compact	3	2	2	9	5	1024 (-5.9%)	-194.1
Gosset-KEM-1024-Compact	4	2	2	10	4	1408 (-10.2%)	-215.0
Gosset-KEM-512-MaxSec	2	4	3	10	4	768	-146.6
Gosset-KEM-768-MaxSec	3	3	2	10	4	1088	-166.8
Gosset-KEM-1024-MaxSec	4	3	2	11	5	1568	-189.4

Compact sets keep the ML-KEM noise, so they keep its lattice security. MaxSec sets keep FIPS 203 sizes and use more noise ($\eta = 4$ is a straightforward extension of the CBD); among all choices with FIPS ciphertext size and a DFR no worse than FIPS, they are optimal.

Lattice security. We evaluated both LWE instances, the key $(\mathbf{s}, \mathbf{e})$ and the ciphertext ($\mathbf{r}$ as secret, $\mathbf{e}_1$ plus optional rounding as error), with the lattice-estimator [9] (all attacks except Arora–GB and BKW, MATZOV cost model [10]). The weakest link, in $\log_2$ of bit operations:

	512	768	1024
ML-KEM	139.7	196.4	262.3
Gosset-KEM MaxSec	144.9	207.2	271.7
gain (rounding credited / not credited)	+5.2 / +5.6	+10.8 / +4.8	+9.4 / +5.3

The dual-hybrid attack is the cheapest in every case. At levels 768 and 1024 the ciphertext instance becomes the bottleneck, because MaxSec there raises η_1 but not η_2.

6 Correlation-based failure boosting

6.1 The effect

For two coefficients a, b of the same ciphertext,

$$\mathrm{Cov}(\delta_a, \delta_b) = \sigma_S^2\, A(b-a), \qquad A(d) = \sum_{\text{polys}} \sum_{j} c_j c_{j+d} \text{ (negacyclic)},$$

so the covariance depends on the attacker-known autocorrelation of C. A standard decoder fails when a single coefficient is too large, so these covariances do not matter to it. An E_8 decoder fails when $\langle \delta, u \rangle > q/2$ for a relevant u. The variance of that projection is $\sigma_S^2 u^T M u = 2\sigma_S^2 T \cdot R(u)$, with $T = \|C\|^2$ and

$$R(u) = 1 + \sum_{a<b} \frac{2u_a u_b}{\|u\|^2} \rho_{d_{ab}}, \qquad \rho_d = A(d)/T.$$

A random ciphertext has $\rho_d \approx \mathcal{N}(0, 1/(2kn))$. An adversary who grinds messages m (the CCA setting: C and G are deterministic functions of m) can select ciphertexts with large $R(u)$ for some u. This multiplies the failure exponent by roughly $1/R$.

Observation 2 (Consecutive blocks amplify the effect)**.** *With blocks of 8 consecutive coefficients, index distance d appears in $8-d$ pairs of every block. For the all-plus type-$(\frac{1}{2}^8)$ vector, the coefficient of ρ_1 in R is $7/4$, and the standard deviation of $R-1$ is $\sqrt{\sum_{d=1}^{7}((8-d)/4)^2} \approx 2.96$ times that of a single ρ_d. Every block shares the same distances, so one ground ciphertext boosts all 32 blocks at once.*

6.2 Tail model

$Z = C^T Q_u C / C^T C$ is a ratio of quadratic forms. Negacyclic shifts are diagonalised by the odd Fourier basis, so $Q_u = \sum_\ell w_\ell (P^\ell + P^{-\ell})/2$ has eigenvalues $\mu_j = \sum_\ell w_\ell \cos(\pi(2j+1)\ell/n)$. Then $\Pr[Z \geq x] = \Pr[C^T(Q_u - xI)C \geq 0]$.

- A Chernoff bound with the Gaussian MGF of this form overestimates Monte Carlo tails by a factor of 7–10, so it favours the attacker.
- The Lugannani–Rice saddlepoint approximation matches Monte Carlo (with the true CBD and rounding entries) within 2–15% down to 10^{-4}, for both single directions and the union over all directions.

- The tails are visibly heavier than Gaussian, so a naive Gaussian model on ρ underestimates the attack.

We combine this with norm grinding [6], using a Chernoff bound on $\Pr[T \geq t]$. The per-ciphertext failure model was validated on $8 \cdot 10^4$ real encryptions at inflated noise: 299 failures observed against 303 predicted, consistent within each correlation bucket.

6.3 Attack cost and countermeasure

The cost of the first failure is $W = 1/(\alpha\beta)$ with $Q = 1/\beta \leq 2^{64}$ queries, where α is the fraction of usable ciphertexts and β their failure probability.

Countermeasure. We use a fixed public permutation `E8_LAYOUT` that assigns 8 coefficient positions to each block such that the 28 pairwise distance classes $\min(d, 256 - d)$ inside every block are distinct. Such a layout is easy to find by randomised greedy search; ours uses each distance class in at most 9 blocks. Now a single large ρ_d affects one pair per block, with weight $\frac{1}{4}$ in the type-$(\frac{1}{2}^8)$ directions, and the standard-deviation multiplier drops from 2.96 to 1.32. Distance 128 has zero correlation, since $A(256 - d) = -A(d)$. The average DFR is unchanged: in a real-encryption test both layouts gave $2.11 \cdot 10^{-5}$, with disjoint failing ciphertexts.

$\log_2$ work, first failure ($Q \leq 2^{64}$)	consecutive	**distinct layout**	claimed level
Gosset-KEM-512-Compact	127.4	**192.0**	≈140
Gosset-KEM-512-MaxSec	147.6	**232.4**	144.9
Gosset-KEM-768-Compact	249.0	**365.4**	≈196
Gosset-KEM-768-MaxSec	242.2	**386.6**	207.2
Gosset-KEM-1024-Compact	554.4	**790.7**	≈262
Gosset-KEM-1024-MaxSec	336.4	**552.3**	271.7

With norm grinding alone the cost is $\geq 2^{515}$ for every set, ML-KEM included, so the gap comes entirely from correlation. After 2^{64} grinding, the best per-ciphertext failure probability for Gosset-KEM-512-MaxSec is $2^{-92.8}$ with the layout and $2^{-70.9}$ without; ML-KEM-512 reaches $2^{-103.7}$.

6.4 After the first failure

Using the geometric model of [7] with attacker-favourable assumptions (the failing direction is known, and each failure carries full information), later failures become much cheaper in every scheme. They remain costlier for Gosset-KEM than for ML-KEM: the second failure costs 2^{145} against 2^{85} at level 512 and 2^{200} against 2^{103} at level 768. The larger relative threshold of E_8 outweighs its larger number of decoding directions ($32 \cdot 240$ against 256). The first failure therefore dominates, as for Kyber [7]. Multi-target attacks are not counted, as for ML-KEM. With the layout, Gosset-KEM-512-Compact tolerates up to 2^{49} targets before falling below 2^{143}.

6.5 Higher dimensions: the Leech lattice

For Λ_{24} (10 blocks of 24 coordinates), the minimal vectors of type $(\mp 3, \pm 1^{23})$ involve 276 pairs. Only 127 distance classes exist, so no layout makes them distinct. The worst-direction multiplier is 4.90 with consecutive blocks and 2.14 with the best layout we found, against 1.32 for E_8. Larger codes buy more average-case margin but are more exposed to this attack. *Any multi-dimensional code for LWE message encoding should be analysed against correlation-based boosting, and its coefficient layout chosen accordingly.*

7 Implementation

Correctness. The implementation is C99 with AVX2. It passes the NIST ACVP vectors for ML-KEM (180/180, plus 60/60 key checks), and a separate reference build agrees bit for bit on all nine parameter sets. A deterministic cross-platform digest over all schemes is identical under gcc and clang on Linux and on Windows (MinGW).

Constant time. The decoder is branch-free; an AVX2 version runs in about 1,300 cycles, against about 300 for the reference `poly_tomsg`. The decapsulation overhead is at most 0.9% for Compact and 2–7% for MaxSec, the latter mostly from extra SHAKE output. dudect-style Welch tests stay at $|t| \leq 3.5$ on Intel Xeon and AMD Zen 3, while positive controls reach $|t| > 10^4$. Two findings:

- **Compiler-inserted branches.** clang 18 `-O3` turned the mask logic of a portable reference decoder into secret-dependent branches ($|t| \approx 1700$); gcc did the same to a Leech decoder. Value barriers fix both, and timing must be checked for each compiler.
- **A fixed-input artefact on Zen 3.** One gcc binary showed $|t|$ of 10 to 19 at 10^6 measurements, only when the same ciphertext was repeated. The effect was 5–10 cycles out of 85,000. ML-KEM-512 showed it too, and valid and invalid ciphertexts remained indistinguishable ($|t| \leq 1.9$). We classify it as a low-impact microarchitectural effect on public input.

Masking. We built a first-order masked decapsulation. Masked Keccak uses ISW AND gadgets [12]; the CBD is computed on bits with B2A conversion; the ciphertext comparison uses public decompression intervals; and the key is selected on shares, so only the chosen key is unmasked. A multiplication-free reformulation of the E_8 decision, $D = \sum a_i - 4q + p_A(2q - 2a_{\max}) - p_B\, 2a_{\min}$, allows a bitsliced masked decoder with 32 blocks per word: 34k cycles, 12.3 times faster than per-block masking. The full masked decapsulation costs 4.7–6.4 times the unmasked one. In simulated TVLA (Hamming-weight model, $\sigma = 1$, 43,956 points, up to $3 \cdot 10^4$ traces per class, four runs), the first-order $|t|$ stays below the Bonferroni threshold or does not repeat across runs, while second-order leakage remains, as expected.

Hybrid and DEM. An X-Wing-style combiner [13] with X25519 gives Mode C (ML-KEM-768, FIPS-compliant) and Mode S (Gosset-KEM-768-MaxSec). A DEM with SHAKE-derived keys, ISO 7816-4 padding and encrypt-then-MAC (HMAC-SHA3-256) completes the hybrid encryption.

8 Limitations

- Gosset-KEM is not FIPS 203 and is not certifiable as it stands. A compliant deployment uses Mode C, or a three-component hybrid.
- The DFR and failure-boosting analyses rely on Gaussian conditional models, validated empirically down to about 10^{-5} per block. Beyond that they are extrapolations, as for ML-KEM.
- The masking results come from simulation; hardware validation, a production CSPRNG and higher-order masking remain open.
- Key generation does not bind the parameter set: $G(d\|k)$ is the same for ML-KEM-512 and Gosset-KEM-512-Compact, so one seed yields identical keys in both. Key pairs must never be reused across sets.
- Independent cryptanalysis is needed.

9 Conclusion

E_8 message coding is an effective way to buy margin in ML-KEM, either as smaller ciphertexts or as +5 to +11 bits of lattice security at FIPS sizes. Its multi-dimensional decoder, however, introduces an attack surface the standard decoder does not have: correlated decryption errors that an adversary can select for. With a naive consecutive layout this pushes the level-1 Compact set below its claimed security. A distinct-distance coefficient layout removes most of the effect at no cost. Higher-dimensional codes are inherently more exposed. We recommend that future proposals of lattice codes for LWE-based KEMs include this analysis.

Availability. Code, specification, test vectors and all analysis scripts: `https://github.com/calabroadrian/gosset-kem`.

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References

[1] NIST. *FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard*. 2024.

[2] S. Liu, A. Sakzad. *On the Maximum Toroidal Distance Code for Lattice-Based Public-Key Cryptography*. arXiv:2601.08452, 2026.

[3] S. Liu, A. Sakzad. *Lattice Codes for CRYSTALS-Kyber*. arXiv:2308.13981, 2023.

[4] C. Saliba, L. Luzzi, C. Ling. *Error Correction for FrodoKEM Using the Gosset Lattice*. arXiv:2110.01740, 2021.

[5] T. B. Paiva, M. A. Simplicio Jr, S. M. Hafiz, B. Yildiz, E. L. Cominetti, H. S. Ogawa. *Tailorable codes for lattice-based KEMs with applications to compact ML-KEM instantiations*. IACR TCHES 2025; ePrint 2024/1243.

[6] J.-P. D'Anvers, Q. Guo, T. Johansson, A. Nilsson, F. Vercauteren, I. Verbauwhede. *Decryption failure attacks on IND-CCA secure lattice-based schemes*. PKC 2019.

[7] J.-P. D'Anvers, M. Rossi, F. Virdia. *(One) failure is not an option: Bootstrapping the search for failures in lattice-based encryption schemes*. EUROCRYPT 2020; ePrint 2019/1399.

[8] D. Hofheinz, K. Hövelmanns, E. Kiltz. *A modular analysis of the Fujisaki–Okamoto transformation*. TCC 2017.

[9] M. R. Albrecht et al. *lattice-estimator*. `https://github.com/malb/lattice-estimator`, commit f31b86f.

[10] MATZOV. *Report on the Security of LWE: Improved Dual Lattice Attack*. 2022.

[11] J. H. Conway, N. J. A. Sloane. *Sphere Packings, Lattices and Groups*. Springer, 3rd ed., 1999.

[12] Y. Ishai, A. Sahai, D. Wagner. *Private circuits: securing hardware against probing attacks*. CRYPTO 2003.

[13] M. Barbosa, D. Connolly, J. D. Duarte, A. Kaiser, P. Schwabe, K. Varner, B. Westerbaan. *X-Wing: The Hybrid KEM You've Been Looking For*. ePrint 2024/039.